A bidirectional neural interaction between syntax and prosody in continuous speech processing

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To understand the structure and meaning of spoken language, the human brain relies on prosody. Although the relationship between syntax and prosody is both formally and experimentally well-established [1,2], how these two linguistic systems interact in the brain, especially under naturalistic listening conditions, remains poorly understood. To address this gap, we analyzed neural responses recorded during audiobook listening to test both how prosody modulates the neural encoding of syntactic structure and how knowledge of syntax shapes the encoding of prosody.

For each word in 49 minutes of Dutch audiobook stories, we computed two syntactic features: the number of phrases that were opened ('node opening') and closed (or integrated; 'node closure'; Fig. 1a). Moreover, we derived a word-based measure of prosodic boundary strength using the Wavelet Prosody Toolkit (Fig. 1b; [3]). Analyses of these auditory stimuli revealed a probabilistic relationship between prosody and syntax in speech: words with strong prosodic boundaries are primarily phrase-closing words, whereas weakly marked words are generally phrase-opening words (Fig. 1e-f). Moreover, boundary strength was positively correlated with the number of nodes that are closed at each word (Fig. 1c). Thus, prosody probabilistically cues syntactic function, which has implications for language processing: a speaker's prosody can inform the listener when to integrate a phrase, and their syntax indicates when prosody will be informative for structure building. We tested these two hypotheses in a magnetoencephalography (MEG) experiment.

Twenty-six Dutch-speaking participants listened to the audiobooks while their brain activity was measured with MEG. We used time-resolved linear regression to map the prosodic and the two syntactic features onto sensor-level MEG data, while controlling for the acoustic and statistical properties of the stimuli by explicitly modeling several additional features (i.e., acoustic spectrogram, acoustic onsets, word onset, word frequency, word surprisal). To test for the interaction between syntax and prosody, we marked each word in terms of its syntactic function (phrase-opening vs. phrase-closing) and in terms of having a strong or a weak boundary. The variance explained by each feature was quantified via hierarchical model comparison.

Both syntactic features explain unique variance over bilateral frontal-temporal sensors, consistent with neural encoding of syntax across time [4,5]. The prosodic feature instead did not improve model fit, suggesting that there might not be a stable, unitary response to prosody. Indeed, the neural response to prosodic boundary strength was modulated by syntax: it was significant for phrase-closing but not for phrase-opening words. This indicates that the brain responds to prosody in a heavily context-dependent way: the extent to which it encodes prosodic features of speech depends on the potential relevance of these features for broader goals of comprehension – i.e., establishing a structured meaning in context. Conversely, and in line with recent work [6], prosody also modulated the neural encoding of syntax: node closure was predictive of brain activity only for words with strong boundary marking, whereas the effect of node opening was significant only for words with weak prosodic boundaries.

In sum, naturally produced speech features a probabilistic relationship between syntactic phrase closure and prosodic boundary marking. Our MEG results show that the brain utilizes this statistical correspondence for the processing of both syntax and prosody: the brain attempts syntactic integration (represented by node closure) primarily when a word is prosodically strongly marked, and it weights prosodic marking especially heavily when this marking can be syntactically informative.

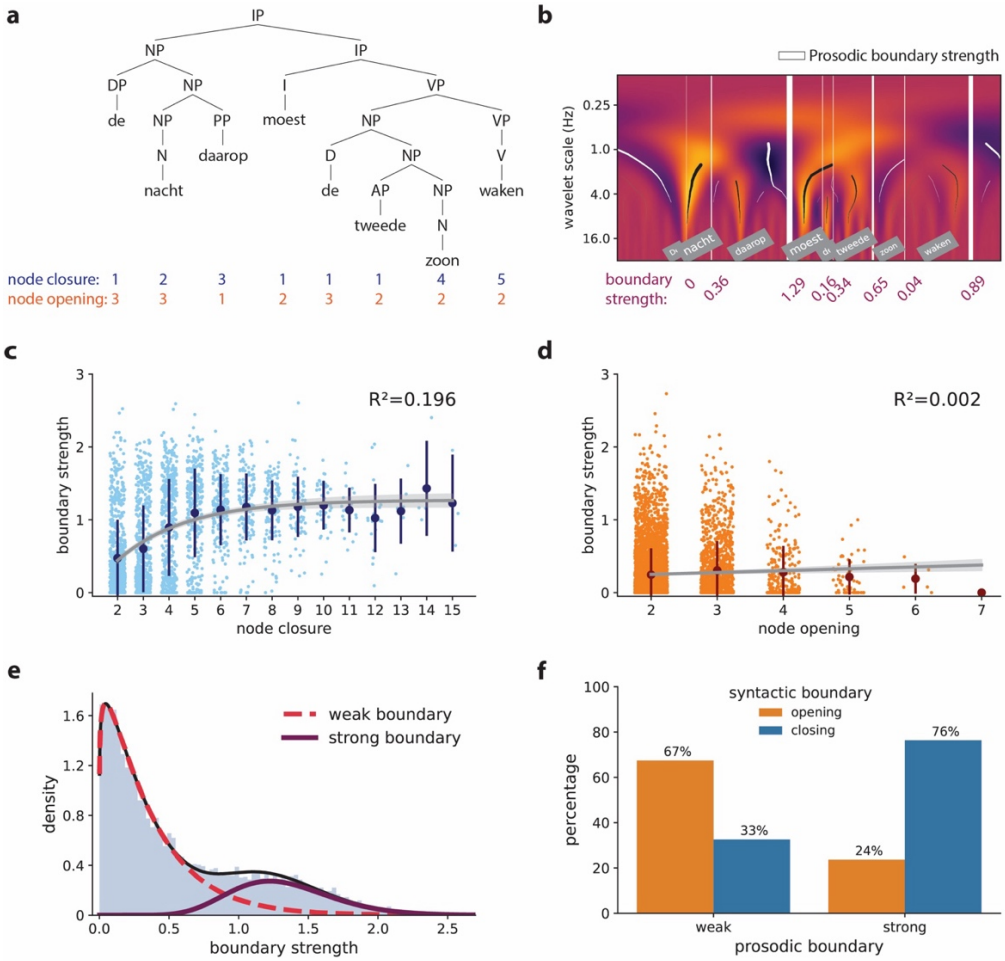

Figure 1. Stimulus annotations and analyses. (a) shows the syntactic structure of an example sentence, (b) shows the prosodic wavelet analysis of that same sentence. (c)-(f) show the correlation between syntactic and prosodic features across all audiobook stories.

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